set_os

Command: `set_os`

Description

Command issued to set new firmware & initiate firmware update

Request

Request Fields

Field	Type	Required	Description	Example
`command`	string	Yes	Specifies the operation being performed, in this case, setting the reader operating system.	`set_os`
`requestId`	string	Yes	A unique identifier for the request, allowing tracking and debugging of the operation.	`abc123`
`OSUpdateDetails`	object	No		
`OSUpdateDetails.url`	string	Yes	The URL where the OS update can be downloaded. Must be a valid URI.	`https://192.168.29.39:8000/Build-3.10.27/Firmware`

Field	Type	Required	Description	Example
`OSUpdateDetails.authenticationType`	string (enum)	Yes	Specifies the type of authentication required for accessing the URL. NONE - No authentication is needed. BASIC - Username and password are required. CERTIFICATE - A valid inline certificate or preinstalled certificate is required. TOKEN - Token-based authentication is required.	`BASIC`
`OSUpdateDetails.authOptions`	object	No	Contains authentication options such as username and password for BASIC authentication.	
`OSUpdateDetails.authOptions.username`	string	No	The username for BASIC authentication.	`admin`
`OSUpdateDetails.authOptions.password`	string	No	The password for BASIC authentication.	`password`
`OSUpdateDetails.headerParam`	array	No	A list of custom header parameters to include in the HTTP request.	
`OSUpdateDetails.queryParam`	array	No	A list of query parameters to include in the URL.	

Field	Type	Required	Description	Example
`OSUpdateDetails.verificationType`	string (enum)	No	Specifies the type of verification to perform during the connection. NONE - No verification. VERIFY_PEER - Verify the peer's certificate. VERIFY_HOST - Verify the host's certificate. VERIFY_HOST_PEER - Verify both the host and peer certificates.	`NONE`
`OSUpdateDetails.caCertificateFileContent`	string	No	The content of the CA certificate in PEM format, required for certificate-based authentication.	`-----BEGIN CERTIFICATE-----\\r\\nPUdg=\\r\\n-----END CERTIFICATE-----`
`OSUpdateDetails.caCertificateFile`	string	No	The file path to the CA certificate, required for certificate-based authentication.	`/apps/ca.crt`

Request Examples

Firmware update 1

```
{
  "command": "set_os",
  "requestId": "abc123",
  "OSUpdateDetails": {
    "url": "https://192.168.29.39:8000/Build-3.10.27/Firmware",
    "authenticationType": "CERTIFICATE",
    "verificationType": "VERIFY_HOST",
    "caCertificateFileContent": "-----BEGIN CERTIFICATE-----\\\\r\\\\nPUdg=\\\\r\\\\n-----END CERTIFICATE-----"
  }
}
```

Firmware update 2

```
{
  "command": "set_os",
  "requestId": "abc123",
  "OSUpdateDetails": {
    "url": "https://192.168.29.39:8000/Build-3.10.27/Firmware",
    "authenticationType": "CERTIFICATE",
    "verificationType": "VERIFY_HOST"
  }
}
```

Firmware update 3

```
{
  "command": "set_os",
  "requestId": "abc123",
  "OSUpdateDetails": {
    "url": "https://192.168.29.39:8000/Build-3.10.27/Firmware",
    "authenticationType": "NONE",
    "verificationType": "NONE"
  }
}
```

Response

Response Fields

Field	Type	Required	Description	Example
`command`	string	Yes		`set_os`
`requestId`	string	Yes		`abc123`
`apiVersion`	string (enum)	Yes		`V1.1`
`response`	object	No	Status of the command execution — check `code` to confirm success (0) or identify the error.	

Field	Type	Required	Description	Example
`response.code`	integer	Yes	Response code indicating success or failure. • **0** — Success • **1** — Command payload accepted • **2** — Invalid payload • **3** — Not able to retrieve information • **4** — Firmware update in progress • **5** — Can't reboot, inventory in progress • **6** — Region is not configured • **7** — Interface is not available • **8** — Insufficient flash size • **9** — File not found • **10** — Configuration already exists • **11** — Inventory in progress • **12** — No radio operation in progress • **13** — Firmware update failed • **14** — Battery too low for firmware update	`0`

Field	Type	Required	Description	Example
			• **15** — WIFI: SSID not found • **16** — WIFI: Cannot delete active SSID • **17** — WIFI: SSID missing • **18** — WIFI: SSID already exists • **19** — WIFI: SSID count overflow • **20** — WiFi is not supported • **21** — Certificate not found • **22** — Advanced configuration not set • **23** — Invalid enum value • **24** — Max 32 prefilters limit exceeded • **25** — Max 3 publish topics exceeded • **26** — Max 1 subscribe topic exceeded • **27** — Invalid tenant ID length • **28** — Tag match pattern length exceeded	

Field	Type	Required	Description	Example
`response.description`	string	Yes	Human-readable description of the response code (e.g. "Success", "Invalid payload").	`Success`

Response Examples

example1

```
{
  "command": "set_os",
  "requestId": "18996",
  "apiVersion": "V1.1",
  "response": {
    "code": 1,
    "description": "Command payload is accepted"
  }
}
```